Monthly Internship Report

Name: Aashi Pradeep Panchal
Prn: 1032233482
Duration: 6 Months
Day & Date: Wednesday 02/07/2025 - Wednesday 31/12/2025
Company: **Pune Instrumentation Pvt. Ltd. (PIPL)**

MONTHLY PROGRESS REPORT

MONTH – 1 (JULY 2025)
During the first month of my internship, I underwent induction and orientation at Pune Instrumentation Pvt. Ltd., where I was introduced to the company's structure, work domains, safety protocols, and engineering practices. I received exposure to consultancy, modelling, simulation, prototyping, testing, and advanced manufacturing activities carried out at PIPL. I was trained on **Ki CAD**, where I learned schematic design, PCB layout, routing, footprint assignment, and design rule checks. Applying this knowledge, I designed a basic electronic circuit and later fabricated a PCB manually. This involved drilling, soldering, component placement, and microcontroller integration. I faced challenges during code flashing and hardware debugging, which helped me understand practical troubleshooting techniques. Towards the end of the month, I began understanding project requirements for an industrial control panel, including enclosure selection, component identification, and layout planning.
MONTH – 2 (AUG 2025)
In the second month, I worked extensively on the **Hot Well Level Water Controller Panel**. My responsibilities included enclosure preparation, marking and cutouts, component procurement, front panel assembly, PCB integration, and internal wiring. I followed schematic diagrams carefully and practiced neat wiring using proper ferrules, labels, and routing methods. I actively participated in **Acceptance Test Procedures (ATP)**, performing pre-power checks, initial power-up, functional simulations, and iterative testing. Multiple testing cycles helped me improve my understanding of fault identification and correction. I was also involved in third-party inspection testing, where the panel functionality was demonstrated under official supervision. Alongside panel work, I was introduced to the **Bio-Disinfectant Unit (Smart Air Purifier)** project and studied its structure, airflow, enclosure design, and control components.
MONTH – 3 (SEPT 2025)
During this month, my focus shifted primarily to the **Bio-Disinfectant Unit (BDU)** project. I worked on integrating sensors with the **ESP32 microcontroller**, understanding sensor power requirements, pin configurations, and real-time data monitoring. This provided hands-on experience in embedded systems and sensor interfacing. I also developed a system logic **flowchart using MATLAB**, defining decision-making sequences and system responses based on sensor data. Additionally, I studied different **data storage approaches**, comparing local and cloud-based solutions and proposing a basic hybrid data logging model. This month helped strengthen my understanding of system-level design, data handling, and microcontroller-based applications. This marked a significant milestone in terms of both confidence and capability in hardware development.

MONTH – 4 (OCT 2025)
In October, I gained deeper exposure to **industrial control panels**, working closely with engineers during design, assembly, wiring, programming, and testing stages. I assisted in preparing and studying electrical drawings such as single-line diagrams, wiring diagrams, and component layouts. I participated in **panel assembly and component placement**, mounting PLC modules, relays, contactors, power supplies, and terminal blocks as per approved drawings. This phase improved my understanding of industrial layout standards, safety clearances, and maintenance considerations.
MONTH – 5 (NOV 2025)
During this month, I was actively involved in **panel wiring and testing activities**. My work included control wiring, power wiring, PLC I/O wiring, and terminal tightening under mentor supervision. I assisted in PLC input-output checking, signal verification, and identifying minor wiring issues. I supported **panel testing and debugging**, including power-up observation, indicator behaviour analysis, and troubleshooting. This month gave me a complete understanding of the **end-to-end execution cycle** of industrial control panels, from design interpretation to testing and verification.
MONTH – 6 (DEC 2025)
In the final month of my internship, I focused on **electrical drawings, documentation, and final verification activities**. I assisted in reviewing single-line diagrams, wiring diagrams, and component layouts, ensuring consistency between documentation and physical panels. I also supported **component placement checks, final wiring inspection, and basic verification activities** under supervision. This phase helped reinforce the importance of documentation accuracy and systematic checks before panel dispatch. The internship concluded with reflection on the skills gained, including industrial panel design, assembly, wiring, testing, embedded systems exposure, and professional documentation practices.

CONCLUSION

The internship at **Pune Instrumentation Pvt. Ltd.** provided valuable hands-on experience in industrial control panels, PCB design, embedded systems, testing, and documentation. Working on real projects helped me apply theoretical knowledge to practical engineering tasks while improving my technical accuracy and problem-solving skills. The guidance from experienced engineers enhanced my understanding of industrial standards and professional work practices. Overall, this internship strengthened my technical foundation and prepared me for future roles in the field of electronics and automation.

Mr. Vineesh Mrs. Prasanna Ms. Aashi